EC304 — Network Theory

Semester III • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Fourier Analysis and Three-Phase Unbalanced Systems

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Fourier Series Analysis in Electrical Networks

Definition: Trigonometric and Exponential Fourier Series

Any physically realizable periodic electrical signal $f(t) = f(t+T)$ with fundamental period T and fundamental angular frequency $\omega_0 = 2\pi/T$ satisfying the Dirichlet conditions can be decomposed into an infinite sum of harmonically related sinusoids:

1. **Trigonometric Fourier Series (TFS):**

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \tag{1}$$

where the Fourier coefficients are defined over any arbitrary period t_0 to $t_0 + T$:

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)dt, \quad a_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\cos(n\omega_0 t)dt, \quad b_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\sin(n\omega_0 t)dt \tag{2}$$

2. **Compact (Harmonic) Trigonometric Form:**

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t - \theta_n), \quad c_0 = a_0, \quad c_n = \sqrt{a_n^2 + b_n^2}, \quad \theta_n = \tan^{-1}\left(\frac{b_n}{a_n}\right) \tag{3}$$

3. **Exponential Fourier Series (EFS):**

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \qquad C_n = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)e^{-jn\omega_0 t}dt \tag{4}$$

1.1 Inter-relationship Between Coefficients

The complex exponential and real trigonometric coefficients relate directly via Euler's identities:

$$C_0 = a_0, \qquad C_n = \frac{a_n - jb_n}{2} = \frac{c_n}{2}e^{-j\theta_n}, \qquad C_{-n} = C_n^* = \frac{a_n + jb_n}{2} \tag{5}$$

Conversely:

$$a_n = C_n + C_{-n} = 2\text{Re}\{C_n\}, \qquad b_n = j(C_n - C_{-n}) = -2\text{Im}\{C_n\} \tag{6}$$

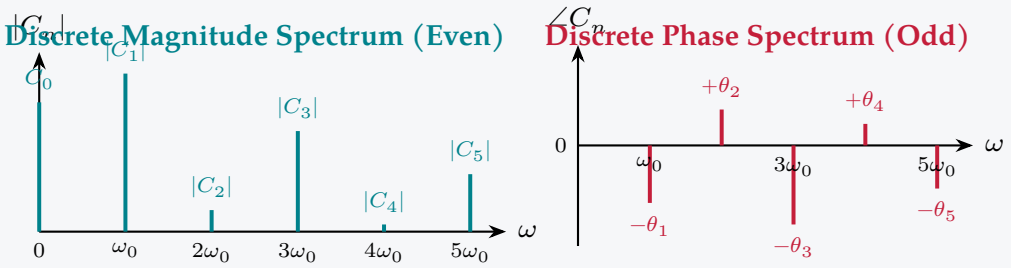
1.2 Waveform Symmetry and Harmonic Content

Exploiting waveform symmetry simplifies Fourier integration and provides immediate diagnostic criteria for harmonic absence:

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Symmetry Type	Mathematical Condition	Zero Coefficients	Surviving Harmonics
Even Symmetry	$f(-t) = f(t)$	$b_n = 0 \quad \forall n$	DC component and cosine harmonics only.
Odd Symmetry	$f(-t) = -f(t)$	$a_0 = 0, \quad a_n = 0 \quad \forall n$	Sine harmonics only (strictly no DC).
Half-Wave Symmetry (HWS)	$f\left(t \pm \frac{T}{2}\right) = -f(t)$	$a_0 = 0, \quad a_{2k} = b_{2k} = 0$	Strictly odd harmonics only ($n = 1, 3, 5, \ldots$).
Even Quarter-Wave	**HWS + Even**	$a_0 = 0, \quad b_n = 0 \quad \forall n$	Odd cosine harmonics only (a_{2k+1}).
Odd Quarter-Wave	**HWS + Odd**	$a_0 = 0, \quad a_n = 0 \quad \forall n$	Odd sine harmonics only (b_{2k+1}).

TikZ Illustration: Discrete Line Spectra of a Periodic Signal



2. Network Response to Non-Sinusoidal Periodic Inputs

2.1 Harmonic Superposition Method

When a linear time-invariant circuit is excited by a non-sinusoidal periodic voltage source:

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_{mn} \cos(n\omega_0 t + \theta_{vn}) \tag{7}$$

the steady-state current response is calculated by applying the **Superposition Principle** to each harmonic component independently.

Systematic Harmonic Solution Procedure

1. **DC Analysis ($n = 0$):** Replace inductors with short circuits ($\mathbf{Z}_L = 0$) and capacitors with open circuits ($\mathbf{Z}_C = \infty$). Compute DC current:
$$I_0 = \frac{V_0}{R_{\text{eq}}} \tag{8}$$
2. **n-th Harmonic Impedance:** Evaluate network input impedance at harmonic frequency $\omega_n = n\omega_0$:
$$\mathbf{Z}(jn\omega_0) = R_n + jX_n = |\mathbf{Z}_n|e^{j\phi_n} = |\mathbf{Z}_n|\angle\phi_n \tag{9}$$

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where inductive reactance scales directly ($X_{Ln} = n\omega_0 L$) and capacitive reactance scales inversely ($X_{Cn} = \frac{1}{n\omega_0 C}$).

3. **n-th Harmonic Current Phasor:**

$$\mathbf{I}_{mn} = \frac{\mathbf{V}_{mn}\angle\theta_{vn}}{|\mathbf{Z}_n|\angle\phi_n} = \frac{V_{mn}}{|\mathbf{Z}_n|}\angle(\theta_{vn} - \phi_n) = I_{mn}\angle\theta_{in} \tag{10}$$

4. **Total Time-Domain Current Synthesis:**

$$i(t) = I_0 + \sum_{n=1}^{\infty} I_{mn}\cos(n\omega_0 t + \theta_{vn} - \phi_n) \tag{11}$$

2.2 Effective (RMS) Values and Average Power

Formulas for RMS Values and Non-Sinusoidal Power

- **True RMS (Effective) Voltage and Current:**

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt} = \sqrt{V_0^2 + \sum_{n=1}^{\infty}\frac{V_{mn}^2}{2}} = \sqrt{V_0^2 + \sum_{n=1}^{\infty} V_{n,\text{rms}}^2} \tag{12}$$

$$I_{\text{rms}} = \sqrt{I_0^2 + \sum_{n=1}^{\infty} I_{n,\text{rms}}^2} \tag{13}$$

- **Total Average Active Power (P):** By orthogonality of distinct sinusoids over fundamental period T, all cross-harmonic terms integrate to zero:

$$P = \frac{1}{T}\int_0^T v(t)i(t)dt = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,\text{rms}} I_{n,\text{rms}}\cos(\phi_n) \tag{14}$$

- **Apparent Power (S), Reactive Power (Q) and Distortion Power (D):**

$$S = V_{\text{rms}} I_{\text{rms}}, \qquad Q = \sum_{n=1}^{\infty} V_{n,\text{rms}} I_{n,\text{rms}}\sin(\phi_n), \qquad D = \sqrt{S^2 - P^2 - Q^2} \tag{15}$$

- **True Power Factor and Distortion Factor:**

$$\text{PF} = \frac{P}{S} = \frac{I_{1,\text{rms}}}{I_{\text{rms}}}\cdot\cos\phi_1 = (\text{Distortion Factor}) \times (\text{Displacement Factor}) \tag{16}$$

- **Total Harmonic Distortion (THD):**

$$\text{THD}_I = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}} = \sqrt{\left(\frac{I_{\text{rms}}}{I_{1,\text{rms}}}\right)^2 - 1} \tag{17}$$

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3. The Continuous Spectrum via Fourier Transform

Continuous Fourier Transform Pair

As period $T \to \infty$, discrete spectral lines merge into a continuous density spectrum:

$$X(j\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \tag{18}$$

$$x(t) = \mathcal{F}^{-1}\{X(j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \tag{19}$$

The **Energy Spectral Density** is $\mathcal{E}(\omega) = |X(j\omega)|^2$. Total signal energy is conserved under the transformation via **Parseval's Energy Theorem**:

$$W = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega \tag{20}$$

4. Three-Phase Unbalanced Circuits and Symmetrical Components

4.1 Symmetrical Components Transformation (Fortescue's Theorem)

In an unbalanced three-phase system, the three unsymmetrical phasor voltages or currents can be decomposed into three balanced symmetrical components:

Fortescue's Theorem and the Complex Operator a

Any set of three unbalanced phasors $\mathbf{V}_a, \mathbf{V}_b, \mathbf{V}_c$ can be uniquely expressed as the sum of three balanced symmetrical sets:

1. **Positive-Sequence ($\mathbf{V}_{a1}, \mathbf{V}_{b1}, \mathbf{V}_{c1}$):** Equal magnitudes, 120° phase separation, same phase sequence (a-b-c) as original.
2. **Negative-Sequence ($\mathbf{V}_{a2}, \mathbf{V}_{b2}, \mathbf{V}_{c2}$):** Equal magnitudes, 120° phase separation, reversed phase sequence (a-c-b).
3. **Zero-Sequence ($\mathbf{V}_{a0}, \mathbf{V}_{b0}, \mathbf{V}_{c0}$):** Equal magnitudes and identical phase angles (in-phase).

Defining the complex phase-shift operator $a = e^{j120^\circ} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$:

$$a^2 = e^{j240^\circ} = -\frac{1}{2} - j\frac{\sqrt{3}}{2}, \qquad a^3 = 1, \qquad 1 + a + a^2 = 0 \tag{21}$$

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Matrix Transformation Relations

The sequence components relate to the phase quantities by:

$$\begin{bmatrix} \mathbf{V}_a \\ \mathbf{V}_b \\ \mathbf{V}_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} \mathbf{V}_{a0} \\ \mathbf{V}_{a1} \\ \mathbf{V}_{a2} \end{bmatrix} = [\mathbf{A}]\mathbf{V}_{012} \tag{22}$$

The inverse transformation matrix $[\mathbf{A}]^{-1}$ yields:

$$\begin{bmatrix} \mathbf{V}_{a0} \\ \mathbf{V}_{a1} \\ \mathbf{V}_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \mathbf{V}_a \\ \mathbf{V}_b \\ \mathbf{V}_c \end{bmatrix} \tag{23}$$

Explicitly:

$$\mathbf{V}_{a0} = \frac{\mathbf{V}_a + \mathbf{V}_b + \mathbf{V}_c}{3}, \quad \mathbf{V}_{a1} = \frac{\mathbf{V}_a + a\mathbf{V}_b + a^2\mathbf{V}_c}{3}, \quad \mathbf{V}_{a2} = \frac{\mathbf{V}_a + a^2\mathbf{V}_b + a\mathbf{V}_c}{3} \tag{24}$$

Physical Consequence: Zero-Sequence Current in Three-Phase Circuits

In a three-phase system, the neutral current $\mathbf{I}_n$ is the sum of line currents:

$$\mathbf{I}_n = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 3\mathbf{I}_{a0} \tag{25}$$

Conclusion:

- In a 3-phase 3-wire system (ungrounded), $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0 \implies \mathbf{I}_{a0} = 0$. Zero-sequence line current **cannot** flow in any 3-wire system.
- In delta-connected loads, line voltages sum to zero ($\mathbf{V}_{ab} + \mathbf{V}_{bc} + \mathbf{V}_{ca} = 0 \implies \mathbf{V}_{L0} = 0$), so zero-sequence voltage cannot exist across lines.

4.2 Neutral Shift Voltage (Millman's Theorem in 3-Phase Systems)

When an unbalanced star-connected load with phase admittances $\mathbf{Y}_a, \mathbf{Y}_b, \mathbf{Y}_c$ is connected to a balanced source with neutral N, the star load neutral point N' floats to a potential known as the **neutral shift voltage** $\mathbf{V}_{N'N}$:

$$\mathbf{V}_{N'N} = \frac{\mathbf{V}_{AN}\mathbf{Y}_a + \mathbf{V}_{BN}\mathbf{Y}_b + \mathbf{V}_{CN}\mathbf{Y}_c}{\mathbf{Y}_a + \mathbf{Y}_b + \mathbf{Y}_c + \mathbf{Y}_n} \tag{26}$$

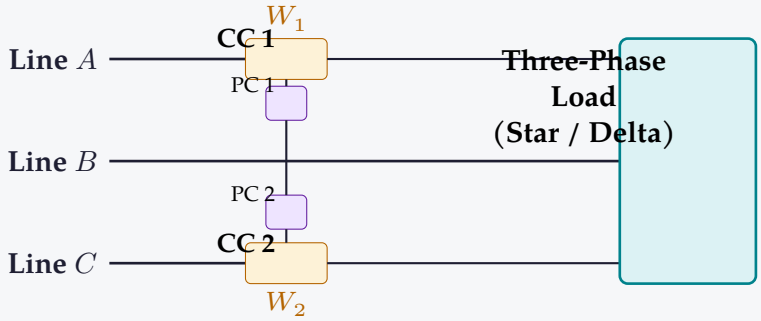
where $\mathbf{Y}_n$ is the neutral wire admittance ($\mathbf{Y}_n = 0$ for 3-wire floating neutral). The voltage across each load branch is then $\mathbf{V}_{aN'} = \mathbf{V}_{AN} - \mathbf{V}_{N'N}$.

4.3 Power Measurement by Two-Wattmeter Method

The two-wattmeter method measures total active power in any 3-phase 3-wire system (balanced or unbalanced, star or delta):

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TikZ Schematic: Two-Wattmeter Connection in a Three-Phase System



Two-Wattmeter Power and Power Factor Relations

Under balanced sinusoidal conditions with line-to-line voltage V_L, line current I_L, and load power factor angle ϕ:

$$W_1 = V_L I_L \cos(30^\circ - \phi), \qquad W_2 = V_L I_L \cos(30^\circ + \phi) \tag{27}$$

1. **Total Active Power (P):**

$$P = W_1 + W_2 = V_L I_L[\cos(30^\circ - \phi) + \cos(30^\circ + \phi)] = \sqrt{3} V_L I_L \cos\phi \tag{28}$$

2. **Total Reactive Power (Q):**

$$W_1 - W_2 = V_L I_L[\cos(30^\circ - \phi) - \cos(30^\circ + \phi)] = V_L I_L \sin\phi \implies Q = \sqrt{3}(W_1 - W_2) \tag{29}$$

3. **Power Factor Angle (ϕ):**

$$\tan\phi = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2} \implies \text{PF} = \cos\phi = \cos\left[\tan^{-1}\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)\right] \tag{30}$$

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Quick Revision — Module 2

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Key Formulas and Network Relations at a Glance

Trigonometric Fourier Form	$f(t) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$
Exponential Fourier Form	$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad C_n = \frac{a_n - jb_n}{2}$
True RMS Voltage	$V_{\text{rms}} = \sqrt{V_0^2 + \sum_{n=1}^{\infty} V_{n,\text{rms}}^2}$
Average Non-Sinusoidal Power	$P = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,\text{rms}} I_{n,\text{rms}} \cos(\phi_n)$
True Power Factor	$\text{PF} = \frac{P}{S} = \frac{I_{1,\text{rms}}}{I_{\text{rms}}} \cos\phi_1$
Total Harmonic Distortion	$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}}$
Complex Operator a	$a = e^{j120^\circ} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}, \quad 1 + a + a^2 = 0$
Zero-Sequence Current	$\mathbf{I}_{a0} = \frac{\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c}{3} = \frac{\mathbf{I}_n}{3}$
Two-Wattmeter Total Power	$P_{\text{total}} = W_1 + W_2 = \sqrt{3} V_L I_L \cos\phi$
Two-Wattmeter Power Factor	$\tan\phi = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$

One-Line Definitions

- **Harmonic Frequency:** An integer multiple ($n\omega_0$) of the fundamental repetition frequency of a periodic wave.
- **Half-Wave Symmetry:** Condition $f(t \pm T/2) = -f(t)$, ensuring complete cancellation of all even harmonic orders.
- **Line Spectrum:** A discrete graphical plot of harmonic component amplitudes and phase shifts versus frequency.
- **Harmonic Impedance:** The frequency-dependent impedance of a circuit evaluated specifically at harmonic frequency $n\omega_0$.
- **Distortion Power (D):** The component of apparent power caused strictly by harmonic cross-frequencies, measured in VA.
- **Displacement Factor:** The cosine of the fundamental phase angle ($\cos\phi_1$), measuring fundamental displacement.
- **Symmetrical Components:** Three balanced phasor sets (positive, negative, zero) replacing an unbalanced three-phase system.
- **Neutral Shift Voltage:** The potential difference appearing between floating load neutral and source star neutral in unbalanced loads.
- **Two-Wattmeter Method:** A standard method using two power meters to measure total power in any three-phase three-wire system.

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- **Parseval's Theorem:** Mathematical relation stating total energy is conserved between time and continuous frequency domains.

Frequently Asked / Viva Questions

Q1. What harmonics are present in a periodic waveform possessing Half-Wave Symmetry?

A waveform with half-wave symmetry ($f(t \pm T/2) = -f(t)$) contains strictly odd harmonics ($n = 1, 3, 5, \dots$). The DC component (a_0) and all even harmonics (a_{2k}, b_{2k}) are identically zero.

Q2. Why does a non-sinusoidal voltage produce an active power contribution only with currents of the identical harmonic order?

Because the time-average product of two sinusoids of differing frequencies over the fundamental period is identically zero: $\int_0^T \cos(m\omega_0 t)\cos(n\omega_0 t)dt = 0$ for $m \neq n$. Active power is generated exclusively by voltage and current terms belonging to the same harmonic frequency.

Q3. Can zero-sequence currents flow in a balanced or unbalanced 3-phase 3-wire system? Explain.

No. In any 3-phase 3-wire system, by KCL, the sum of line currents must be zero ($\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$). Since zero-sequence current is $\mathbf{I}_{a0} = \frac{1}{3}(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c)$, it is strictly zero. A neutral or ground path is required for zero-sequence current to exist.

Q4. What does it mean if one wattmeter reads negative in the two-wattmeter method?

A negative reading occurs when the load power factor is less than 0.5 ($\phi > 60^\circ$). In that case, $W_2 = V_L I_L \cos(30^\circ + \phi)$ becomes negative. Total power is still the algebraic sum $W_1 + W_2$.

Q5. What are the readings of the two wattmeters at unity power factor ($\cos\phi = 1$)?

At unity power factor, $\phi = 0^\circ$. Thus, $W_1 = V_L I_L \cos(30^\circ) = \frac{\sqrt{3}}{2} V_L I_L$ and $W_2 = V_L I_L \cos(30^\circ) = \frac{\sqrt{3}}{2} V_L I_L$. Both wattmeters read identical positive values ($W_1 = W_2 = P/2$).

Q6. What are the readings of the two wattmeters at zero power factor ($\cos\phi = 0$)?

At zero power factor, $\phi = 90^\circ$. Then $W_1 = V_L I_L \cos(-60^\circ) = 0.5 V_L I_L$ and $W_2 = V_L I_L \cos(120^\circ) = -0.5 V_L I_L$. The wattmeters have equal magnitudes but opposite signs ($W_1 = -W_2$), yielding net active power $P = W_1 + W_2 = 0$.

Q7. State the relationship between Total Harmonic Distortion (THD) and Distortion Factor.

The distortion factor $\mu = I_{1,\text{rms}}/I_{\text{rms}}$ relates to THD by $\mu = \frac{1}{\sqrt{1+\text{THD}^2}}$. As harmonic distortion increases, the distortion factor decreases below unity, degrading true power factor.

Q8. What is the physical meaning of the complex operator $a = e^{j120^\circ}$?

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Frequently Asked / Viva Questions (contd.)

The operator a represents a counter-clockwise phase rotation of 120° ($2\pi/3$ radians) without altering the magnitude of the phasor. Multiplying by a^2 rotates by 240°.

Q9. How does Millman's Theorem find the neutral shift in an unbalanced 3-phase star circuit?

Millman's theorem treats the 3 phase voltages as parallel branches connected to neutral. The star point voltage is $\mathbf{V}_{N'N} = (\sum \mathbf{V}_k \mathbf{Y}_k)/\sum \mathbf{Y}_k$, directly yielding the shift in neutral potential.

Q10. Distinguish between continuous and discrete Fourier spectra.

A discrete spectrum consists of isolated impulses at integer multiples of fundamental frequency $n\omega_0$, characteristic of periodic signals. A continuous spectrum is a continuous function of ω spanning all frequencies, characteristic of non-periodic transient energy pulses.

Summary Matrix: Waveform Symmetries and Two-Wattmeter Diagnostic Rules

Operating Condition	Wattmeter Relationship	Power Factor / Significance
$\phi = 0^\circ$ **(Resistive)**	$W_1 = W_2 > 0$	PF $= 1.0$; identical positive readings.
$\phi = 60^\circ$	$W_1 > 0, \quad W_2 = 0$	PF $= 0.5$; meter 2 reads exactly zero.
$\phi > 60^\circ$ **(Lagging)**	$W_1 > 0, \quad W_2 < 0$	PF < 0.5; meter 2 reverses deflection.
$\phi = 90^\circ$ **(Pure Reactive)**	$W_1 = -W_2$	PF $= 0$; net active power is strictly zero.
Unbalanced 3-Wire System	$W_1 + W_2 = P_{\text{total}}$	Valid regardless of load unbalance or power factor.

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